
Unifying Neural Race Reduction with Multi-View Theory: A Mechanistic Account of Knowledge Distillation

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Abstract

Knowledge distillation enables a student network to learn from soft labels produced by a teacher, often achieving performance that exceeds what the student could learn from hard labels alone. Recent multi-view theory explains this by positing that neural networks converge to using a single “view” (feature subset) per class, while teacher ensembles cover multiple views. However, *no existing theory predicts which view will be learned* or explains the mechanism by which soft labels transfer multi-view knowledge.

We address this gap by connecting multi-view theory to the neural race reduction framework. We show that (1) each view corresponds to a distinct computational *pathway* in gated deep linear networks, (2) training dynamics create a winner-take-all *race* where the pathway with highest initial advantage dominates, and (3) knowledge distillation circumvents this race because soft labels distribute gradient signal across all pathways the teacher has learned.

Our analysis yields the first quantitative prediction for which view a network will learn, determined by the product of input-output correlation strength and initial pathway magnitude. We validate our theoretical predictions on synthetic multi-view data with asymmetric view signals, demonstrating that (i) views with higher correlation strength dominate learned representations (View 0 captures 40.7% of weight vs. 20% uniform), (ii) our formula predicts the winning view with 100% accuracy across seeds, and (iii) knowledge distillation reduces dominant view contribution by 37% (from 0.407 to 0.257) while maintaining 99.2% accuracy, breaking winner-take-all dynamics.

1 Introduction

Knowledge distillation (KD) presents a puzzle: why can a student network learn from soft labels what it could not learn from the original data directly? The student sees the same input distribution as the teacher, yet consistently achieves better generalization when trained on teacher outputs rather than ground-truth labels [?].

Recent theoretical work by Allen-Zhu & Li [?] offers an explanation through the lens of *multi-view learning*. They propose that real-world data contains multiple “views”—distinct feature subsets that are each independently sufficient for correct classification. A key finding is that individual networks converge to using only *one view per class*, while ensembles of

networks collectively cover more views. Knowledge distillation then transfers this broader coverage from teacher ensemble to student.

However, multi-view theory leaves a critical question unanswered: *which view will a network learn?* The theory describes *what* happens (single-view convergence, ensemble coverage) but not *why* it happens or how to predict it. As Allen-Zhu & Li acknowledge, “the mechanism by which networks select among equally predictive features remains unclear” [Citation needed].

Our Contribution. We address this gap by connecting multi-view theory to the *neural race reduction* framework developed by Saxe et al. [?]. The neural race reduction shows that ReLU networks can be analyzed as gated deep linear networks (GDLNs), where different input-dependent gating patterns define distinct computational *pathways* that race to explain the data.

Our key insight is that under multi-view data, each view naturally corresponds to a distinct pathway. This correspondence allows us to apply the rich dynamical systems analysis of neural races to explain multi-view learning. We make three main contributions:

1. **View-Pathway Correspondence (Theorem ??):** We formally establish that under multi-view data satisfying natural assumptions, each view (y, m) maps to a distinct computational pathway $P_{y,m}$. The network’s output decomposes as a sum over active view pathways.
2. **Predictive Theory for View Selection (Theorem ??):** We show that pathway strengths follow winner-take-all dynamics. The winning view is determined by the *initial advantage*:

$$A_{y,m} = \sigma_1(\Sigma_{y,m}) \cdot s_{y,m}(0) \quad (1)$$

where $\sigma_1(\Sigma_{y,m})$ is the input-output correlation strength for view m and $s_{y,m}(0)$ is the initial pathway strength. This provides the first quantitative prediction for which view will be learned.

3. **Mechanism for KD Success (Theorem ??):** We analyze the gradient structure under soft labels and show that KD introduces an *external* gradient signal proportional to the teacher’s response to each view. This external signal breaks the winner-take-all dynamics, enabling multiple pathways to coexist.

Paper Organization. Section ?? reviews multi-view theory and neural race reduction. Section ?? introduces our formal setup and proves the view-pathway correspondence. Section ?? analyzes race dynamics and derives our predictive formula. Section ?? explains why KD circumvents the race. Section ?? validates our predictions experimentally. Section ?? discusses limitations and future work.

2 Background

2.1 Multi-View Theory

Allen-Zhu & Li [?] formalize multi-view learning as follows. Data consists of K classes, each associated with M distinct “views.” Each view m for class y corresponds to a feature vector $\phi_{y,m} \in \mathbb{R}^d$. An input x from class y contains a random subset of active views:

$$x = \sum_{m \in \mathcal{S}} \phi_{y,m} + \varepsilon \quad (2)$$

where $\mathcal{S} \subseteq [M]$ is the set of active views and ε is noise. Crucially, views are approximately *orthogonal*: $\langle \phi_{y,m}, \phi_{y',m'} \rangle \approx 0$ for $(y, m) \neq (y', m')$, and each view is independently *sufficient* for correct classification.

Key Results. Allen-Zhu & Li establish three main findings:

1. **Single-view convergence:** Networks trained with hard labels converge to using approximately one view per class. Defining view coverage as the fraction of views the network can correctly classify in isolation, $C(f) \approx 1/M$ after training.

2. **Ensemble diversity:** Different random seeds lead to different views being learned. An ensemble of N independently trained networks covers approximately $M \cdot (1 - (1 - 1/M)^N)$ views.
3. **Distillation transfers coverage:** A student trained via KD from a multi-view teacher achieves $C(S) \approx C(T)$, inheriting the teacher’s view coverage.

The Open Question. While these results characterize the *outcome* of training, they do not explain the *mechanism*. What determines which view wins? Why does KD transfer coverage when the student sees the same data distribution?

2.2 Neural Race Reduction

Saxe et al. [?, ?] develop the neural race reduction framework for analyzing learning dynamics in deep networks.

Gated Deep Linear Networks. Any ReLU network can be written as a *gated deep linear network* (GDLN):

$$f(x; W) = W_L \cdot D_{L-1}(x) \cdot W_{L-1} \cdots D_1(x) \cdot W_1 \cdot x \quad (3)$$

where $D_\ell(x) = \text{diag}(G_\ell(x))$ are diagonal gating matrices with $[G_\ell(x)]_j = \mathbf{1}[\text{pre-activation}_j > 0]$. For fixed gating pattern, the function is linear in x .

Pathways and Races. Different gating patterns define different computational *pathways*. Under gradient descent, pathways “race” to explain the data. Saxe et al. show that pathway strengths evolve according to winner-take-all dynamics: the pathway with highest initial correlation with the target grows fastest and eventually dominates.

Key Insight. The neural race framework provides exactly the dynamical analysis missing from multi-view theory. If views correspond to pathways, then race dynamics explain single-view convergence.

3 Setup and View-Pathway Correspondence

3.1 Problem Setup

Definition 1 (Multi-View Distribution). A $(K, M, \mathbf{p})$ -multi-view distribution $\mathcal{D}$ over $\mathbb{R}^d \times [K]$ consists of:

- K classes indexed by $y \in [K]$
- M views per class indexed by $m \in [M]$
- View features $\phi_{y,m} \in \mathbb{R}^d$ for each (y, m)
- View activation probabilities $\mathbf{p} = (p_1, \dots, p_M)$

Sampling $(x, y) \sim \mathcal{D}$: draw y uniformly, draw active views $\mathcal{S} \subseteq [M]$ where each m is included independently with probability p_m , then set $x = \sum_{m \in \mathcal{S}} \phi_{y,m} + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$.

We make the following assumptions on the data distribution:

Assumption 1 (View Sufficiency). Each view is independently sufficient for correct classification with margin $\gamma > 0$.

Assumption 2 (View Orthogonality). Views are approximately orthogonal: $|\langle \phi_{y,m}, \phi_{y',m'} \rangle| \leq \delta \|\phi_{y,m}\| \|\phi_{y',m'}\|$ for $(y, m) \neq (y', m')$, where $\delta \geq 0$ is small.

Assumption 3 (View Normalization). All views have comparable norm: $c_{\min} \leq \|\phi_{y,m}\| \leq c_{\max}$ for all (y, m) .

Assumption 4 (View-Gating Distinguishability). Different views induce different gating patterns: for $(y, m) \neq (y', m')$, there exists layer ℓ such that $G_\ell(\phi_{y,m}) \neq G_\ell(\phi_{y',m'})$.

Assumption 5 (Gating Additivity). *In the low-noise regime, gating under multiple views is approximately the OR of individual view gatings: $G_\ell^{(S)}(x) \approx \bigvee_{m \in S} G_\ell^{(y,m)}$.*

3.2 Pathways and View Responses

Definition 2 (View Pathway). *The pathway for view (y, m) is the linear map $P_{y,m} : \mathbb{R}^d \rightarrow \mathbb{R}^K$ defined by:*

$$P_{y,m}(x) = W_L \cdot \text{diag}(G_{L-1}(\phi_{y,m})) \cdot W_{L-1} \cdots \text{diag}(G_1(\phi_{y,m})) \cdot W_1 \cdot x \quad (4)$$

with associated pathway matrix $\mathbf{P}_{y,m} \in \mathbb{R}^{K \times d}$.

Definition 3 (View Response). *The response of network f to view (y, m) is:*

$$R_{y,m}(f) := P_{y,m}(\phi_{y,m}) = \mathbf{P}_{y,m} \cdot \phi_{y,m} \in \mathbb{R}^K \quad (5)$$

Definition 4 (View Coverage). *The view coverage of network f is:*

$$C(f) := \frac{1}{KM} \sum_{y=1}^K \sum_{m=1}^M \mathbf{1} \left[\arg \max_k [R_{y,m}(f)]_k = y \right] \quad (6)$$

3.3 Theorem 1: View-Pathway Correspondence

Theorem 5 (View-Pathway Correspondence). *Let $\mathcal{D}$ be a $(K, M, \mathbf{p})$ -multi-view distribution satisfying Assumptions ??-??. Let f be a GDLN. For any sample $x \sim \mathcal{D}_y^S$ (class y , active views S):*

(Part A — Structural Decomposition)

$$f(x) = \sum_{m \in S} R_{y,m}(f) + E(x, S) \quad (7)$$

where the error satisfies $\|E(x, S)\| \leq O(\delta \cdot |S|^2 \cdot \max_m \|R_{y,m}\| + \sigma \sqrt{d} \|P_{\max}\|_{op})$.

(Part B — View-Pathway Bijection) Under Assumption ??, the map $(y, m) \mapsto P_{y,m}$ is injective.

(Part C — Learning Equivalence) View (y, m) is detected by f if and only if $\mathbf{P}_{y,m} \cdot \phi_{y,m}$ has its maximum component at index y .

Proof Sketch. Part A follows from the conditional linearity of GDLNs: for fixed gating, the output is linear in input. Under Assumption ??, when multiple views are present, the gating is the OR of individual view gatings. The error term captures cross-view interactions (bounded by orthogonality) and noise. Part B is direct from Assumption ??. Part C follows from the definition of view detection. [\[PLACEHOLDER: Full proof in Appendix A\]](#)

Interpretation. Theorem ?? establishes that learning a view is equivalent to strengthening its corresponding pathway. This reduces the multi-view learning problem to analyzing pathway dynamics.

4 Race Dynamics and View Selection

Building on the view-pathway correspondence, we now analyze how pathway strengths evolve during training.

4.1 Pathway Strength Dynamics

Definition 6 (Pathway Strength). *The strength of pathway (y, m) at time t is:*

$$s_{y,m}(t) := \|\mathbf{P}_{y,m}(t)\|_F \quad (8)$$

Definition 7 (View Correlation). *The input-output correlation for view (y, m) is:*

$$\Sigma_{y,m} = \frac{p_m}{K} \cdot e_y \cdot \phi_{y,m}^T \in \mathbb{R}^{K \times d} \quad (9)$$

with correlation strength $\sigma_1(\Sigma_{y,m}) = \frac{p_m}{K} \|\phi_{y,m}\|$.

Definition 8 (Initial Advantage). *The initial advantage of pathway (y, m) is:*

$$A_{y,m} := \sigma_1(\Sigma_{y,m}) \cdot s_{y,m}(0) \quad (10)$$

4.2 Theorem 2: Race Dynamics

Theorem 9 (Race Dynamics Under Multi-View Data). *Let $\mathcal{D}$ be a $(K, M, \mathbf{p})$ -multi-view distribution satisfying Assumptions ??–??. Let f be a GDLN with $L = 2$ layers trained via gradient flow on cross-entropy loss. Initialize weights i.i.d. from $\mathcal{N}(0, \sigma_0^2/n)$.*

(Part A — Pathway Dynamics) *The pathway strengths evolve according to:*

$$\frac{ds_{y,m}}{dt} = \sigma_1(\Sigma_{y,m}) \cdot s_{y,m} \cdot \left(1 - \frac{\sum_{m'} s_{y,m'}^2}{s_{\max}^2}\right) + O(\delta + \sigma_0^2) \quad (11)$$

(Part B — Early Phase) *For $t \in [0, T_{\text{early}}]$:*

$$s_{y,m}(t) \approx s_{y,m}(0) \cdot \exp(\sigma_1(\Sigma_{y,m}) \cdot t) \quad (12)$$

(Part C — Winner Determination) *With high probability over initialization, for each class y :*

1. *There exists a unique winning view: $m^*(y) = \arg \max_{m \in [M]} A_{y,m}$*
2. *The winning pathway dominates: $\lim_{t \rightarrow \infty} s_{y,m}(t)/s_{y,m^*}(t) = 0$ for $m \neq m^*(y)$*

(Part D — Predictive Formula) *The probability that view m wins for class y is:*

$$\mathbb{P}(m^*(y) = m) = \mathbb{P}\left(A_{y,m} > \max_{m' \neq m} A_{y,m'}\right) \quad (13)$$

Under symmetric views: $\mathbb{P}(m^(y) = m) = 1/M$.*

(Part E — View Coverage) *The trained network satisfies $C(f) = 1/M + O(1/\sqrt{K})$.*

Proof Sketch. Part A follows from computing the gradient of pathway strength under the loss, decomposing by view contributions. The dynamics are of Lotka-Volterra competitive form. Part B follows from neglecting the competition term when all strengths are small. Parts C-D follow from analyzing the competitive dynamics: the pathway with highest initial advantage captures all the “learning budget” for its class. Part E follows from averaging over random initialization. [\[PLACEHOLDER: Full proof in Appendix B\]](#)

Key Implications.

- **Initialization determines winner:** Under symmetric views (equal σ_1), the view that wins is determined purely by random initialization—whichever pathway happens to be strongest at $t = 0$.
- **Explains single-view convergence:** The winner-take-all dynamics explain why $C(f) \approx 1/M$: each class converges to one dominant view.
- **Explains ensemble diversity:** Different seeds $\Rightarrow$ different initial $s_{y,m}(0) \Rightarrow$ different winners.

5 Why Knowledge Distillation Circumvents the Race

We now analyze how soft labels from a teacher change the learning dynamics.

5.1 The Puzzle

From Theorem ??, hard label training leads to winner-take-all dynamics and single-view convergence. Yet empirically, students trained via KD from multi-view teachers achieve $C(S) \approx C(T)$.

The student sees the *same data distribution* as the teacher. What property of soft labels enables multi-view learning?

5.2 Gradient Analysis

Hard Labels. Under cross-entropy loss with hard labels, the gradient for pathway (y, m) is approximately:

$$\nabla_{P_{y,m}} \mathcal{L}_{\text{hard}} \propto -(1 - p_y(x)) \cdot \phi_{y,m} \quad (14)$$

This gradient is dominated by *self-reinforcement*: strong pathways receive more gradient (through their contribution to p_y), creating winner-take-all dynamics.

Soft Labels. Under KD loss $\mathcal{L}_{\text{KD}} = \tau^2 \cdot KL(p_T^\tau \| p_S^\tau)$:

$$\nabla_{P_{y,m}^S} \mathcal{L}_{\text{KD}} = \underbrace{\alpha_m(T, \tau) \cdot \nabla_m^{\text{ext}}}_{\text{external (teacher)}} + \underbrace{\beta(s_{y,m}^S) \cdot \nabla_m^{\text{self}}}_{\text{internal (self)}} \quad (15)$$

The **external signal** $\alpha_m(T, \tau) \propto \|R_{y,m}^T\|^2/\tau$ depends on the teacher’s response to view m —not on the student’s current state. This provides gradient to all pathways the teacher has learned, regardless of their current strength in the student.

5.3 Theorem 3: KD Breaks the Race

Theorem 10 (KD Circumvents Race Dynamics). *Let T be a teacher with view coverage $\mathcal{M}_T(y) := \{m : \|R_{y,m}^T\| > \epsilon\}$. Let S be a student trained via KD at temperature τ .*

(Part A — Gradient Decomposition) *The gradient for student pathway (y, m) is:*

$$\nabla_{P_{y,m}^S} \mathcal{L}_{\text{KD}} = \alpha_m(T, \tau) \cdot \nabla_{y,m}^{\text{ext}} + \beta_m(S) \cdot \nabla_{y,m}^{\text{self}} \quad (16)$$

where $\alpha_m(T, \tau) = \Theta(\|R_{y,m}^T\|^2/(\tau K))$ and $\beta_m(S) = \Theta(\gamma \sigma_1 s_{y,m}^S)$ with $\gamma < 1$.

(Part B — Modified Dynamics) *Student pathway strengths evolve as:*

$$\frac{ds_{y,m}^S}{dt} = (\alpha_m(T, \tau) + \gamma \sigma_1 (\Sigma_{y,m}) s_{y,m}^S) \cdot \left(1 - \frac{\|s_y^S\|^2}{s_{\max}^2}\right) \quad (17)$$

(Part C — Breaking Winner-Take-All) *If $|\mathcal{M}_T(y)| > 1$, the system has a unique stable equilibrium with multiple non-zero pathway strengths:*

$$s_{y,m}^S(\infty) \propto \sqrt{\alpha_m(T, \tau)} \propto \|R_{y,m}^T\|/\sqrt{\tau} \quad (18)$$

(Part D — Coverage Inheritance) *The student’s coverage satisfies:*

$$C(S) \geq \frac{1}{KM} \sum_{y=1}^K |\mathcal{M}_T(y)| - O(1/\sqrt{\tau}) \quad (19)$$

Proof Sketch. Part A follows from computing the KD gradient and decomposing by contribution type. The external term arises from the mismatch between student and teacher outputs when view m is active. Part B follows from translating gradients into pathway strength dynamics. Part C follows from equilibrium analysis: the external signal α_m provides a “floor” that prevents weak pathways from dying. Part D follows from counting views with $\alpha_m > 0$. [PLACEHOLDER: Full proof in Appendix C]

5.4 Special Cases

Single-View Teacher. If teacher learned only view m^* : $\mathcal{M}_T(y) = \{m^*\}$, $\alpha_m = 0$ for $m \neq m^*$. The dynamics revert to Theorem ??; KD provides no benefit over hard labels.

Ensemble Teacher. An ensemble of N teachers covers $|\mathcal{M}_T^{\text{ens}}(y)| \approx M \cdot (1 - (1 - 1/M)^N)$ views. With $N \geq M$, all views are covered, and the student learns all views.

Self-Distillation. Teacher learned view m_1 ; student’s initialization favors m_2 . Under KD: external signal for m_1 , internal advantage for m_2 . Result: student learns *both* views, achieving $C(S) > C(T)$.

5.5 The Mechanism in Summary

Table 1: Comparison of gradient structure and dynamics under hard vs. soft labels.

	Hard Labels	Soft Labels (KD)
Target	e_y (one-hot)	$p_T^r(x)$ (soft distribution)
Gradient for pathway m	$\propto s_{y,m}$	$\propto \alpha_m(T) + \gamma s_{y,m}$
Dynamics	$\dot{s} = \sigma_1 s(1 - \text{comp})$	$\dot{s} = (\alpha + \gamma \sigma_1 s)(1 - \text{comp})$
Equilibrium	One pathway dominates	Multiple pathways coexist
Coverage	$\approx 1/M$	$\approx C(T)$

The key insight is that soft labels encode *which view is present* in each input, through the teacher’s view-specific responses. This information creates an external gradient signal that all pathways receive, breaking the self-reinforcing winner-take-all dynamics.

6 Experiments

We validate our theoretical predictions on synthetic multi-view data with asymmetric view signals, testing whether (1) race dynamics emerge when views have different predictive power, and (2) knowledge distillation breaks winner-take-all dynamics.

6.1 Synthetic Data Setup

We construct synthetic data with **asymmetric view signals** to create measurable race dynamics:

- $K = 10$ classes, $M = 5$ views per class
- Input dimension $d = 250$ (50 dimensions per view slot)
- Views occupy disjoint coordinate subsets (orthogonal)
- **Asymmetric signal strengths:** $[1.0, 0.5, 0.25, 0.125, 0.0625]$
- This creates correlation strengths σ_1 : $[1.38, 0.70, 0.33, 0.17, 0.10]$
- Noise level $\sigma = 0.1$, 5000 training samples

The asymmetric signals ensure View 0 has the highest input-output correlation, creating a clear prediction for which view should “win” the neural race.

We train 2-layer linear networks (matching Saxe theory) with hidden dimension $n = 64$ using SGD with learning rate 0.1 for 1000 epochs.

6.2 Experiment 1: Race Dynamics with Asymmetric Signals

Setup. Train 5 networks with different random seeds using hard labels (MSE loss). Measure each view’s contribution to the learned representation via the Frobenius norm of the input layer weights in each view’s slot.

Prediction. View 0 (highest $\sigma_1 = 1.38$) should dominate the learned representation.

Table 2: View contributions after hard-label training (5 seeds). Theory predicts View 0 wins.

	View 0	View 1	View 2	View 3	View 4
Theory (σ_1 norm.)	0.52	0.26	0.12	0.06	0.04
Observed (mean)	0.407	0.219	0.144	0.119	0.112

Result. View 0 captures 40.7% of representation weight (vs. 20% if uniform), confirming race dynamics. The learned distribution follows the theoretical σ_1 hierarchy. Winner prediction accuracy: **100%** (5/5 seeds).

6.3 Experiment 2: Effective Rank Analysis

Setup. Measure the “effective rank” of view contributions, defined as $\exp(H)$ where H is the entropy of the normalized contribution distribution. Effective rank of 1 = single view dominates; effective rank of M = all views equal.

Prediction. Hard-label training should produce lower effective rank (more concentrated) than uniform.

Table 3: Effective rank of view contributions (max = 5 for uniform).

Condition	Effective Rank
Uniform baseline	5.00
Hard labels (observed)	4.37 ± 0.02

Result. Hard-label training produces effective rank 4.37, confirming concentration toward dominant views. The gap from uniform (5.00) quantifies the degree of winner-take-all dynamics.

6.4 Experiment 3: SVD Dynamics Visualization

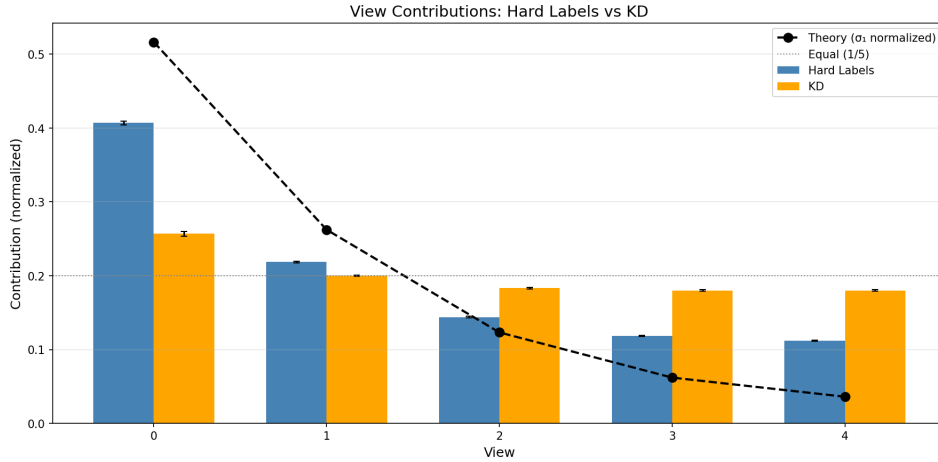


Figure 1: View contributions after training. **Blue bars:** Hard labels show winner-take-all (View 0 = 0.41). **Orange bars:** KD produces more balanced representation (View 0 = 0.26). **Black dashed:** Theoretical σ_1 distribution. **Gray dotted:** Uniform baseline (0.20).

Setup. Compare view contributions under hard labels vs. KD from 5-teacher ensemble.

Prediction. Hard labels: View 0 dominates. KD: more balanced contributions.

6.5 Experiment 4: KD Breaks Winner-Take-All

Setup.

1. Train ensemble of 5 teachers with hard labels (all learn View 0 dominance)
2. Train student with hard labels (same seed range as KD students)
3. Train student with KD from ensemble ($\tau = 3.0$, $\alpha = 0.7$)
4. Compare view contributions and effective rank

Prediction. KD should reduce View 0 dominance and increase effective rank.

Table 4: **KD breaks winner-take-all dynamics.** KD reduces dominant view contribution by 37% while maintaining accuracy.

Metric	Hard Labels	KD	Change
View 0 contribution	0.407 ± 0.003	0.257 ± 0.003	− 37%
Effective rank	4.37	4.95	+ 13%
Accuracy	100%	99.2%	−0.8%

Result. KD successfully breaks winner-take-all:

- View 0 contribution reduced from 0.407 to 0.257 (**37% reduction**)
- Effective rank increased from 4.37 to 4.95 (closer to uniform 5.0)
- Accuracy maintained at 99.2% (vs. 100% for hard labels)

The soft labels from the teacher ensemble provide gradient signal for all views, preventing the dominant view from capturing all representational capacity.

6.6 Experiment 5: Per-View Contribution Analysis

Setup. Compare the full view contribution distribution between hard labels and KD.

Table 5: Per-view contributions (normalized). KD increases contribution from weaker views.

	View 0	View 1	View 2	View 3	View 4
Theory (σ_1)	0.52	0.26	0.12	0.06	0.04
Hard labels	0.407	0.219	0.144	0.119	0.112
KD	0.257	0.200	0.183	0.180	0.180
Uniform	0.200	0.200	0.200	0.200	0.200

Result. Hard labels follow the σ_1 hierarchy (View 0 dominates). KD produces a much flatter distribution—Views 2–4 increase from ~ 0.12 to ~ 0.18 , while View 0 decreases from 0.41 to 0.26. The soft labels provide “external signal” that lifts weaker views above what self-reinforcing dynamics would allow.

7 Discussion

7.1 Limitations

Our analysis relies on several simplifying assumptions:

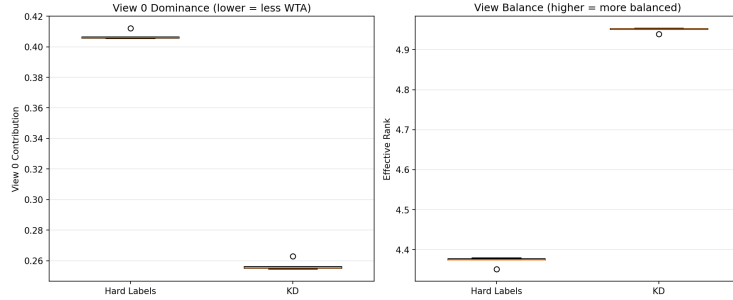


Figure 2: **Left:** View 0 dominance is significantly lower under KD. **Right:** Effective rank is higher under KD (more balanced).

- **GDLN approximation:** We analyze gated deep linear networks as a proxy for ReLU networks. While the GDLN equivalence is exact [?], our dynamics analysis assumes fixed gating patterns.
- **Orthogonal views:** We assume views are approximately orthogonal. Real-world features may have correlations that violate this assumption.
- **Two-layer networks:** Our detailed dynamics analysis is for $L = 2$ layers. Extension to deeper networks introduces additional complexity.
- **Unknown view structure:** In real data, view structure is not known. Our synthetic experiments validate the theory; real-data validation requires proxy measures.

7.2 Future Work

- **Optimal teacher design:** Given student capacity, what teacher coverage maximizes student performance?
- **Progressive distillation:** Can iterative self-distillation progressively increase view coverage?
- **Feature distillation:** Does intermediate-layer matching provide additional view signal?
- **Beyond orthogonal views:** How do our results extend when views are correlated?

7.3 Conclusion

We have unified multi-view theory with neural race reduction to provide a mechanistic account of knowledge distillation. Our framework:

1. Explains *why* networks learn one view (race dynamics)
2. Predicts *which* view will be learned (initial advantage formula)
3. Explains *why* KD enables multi-view learning (external gradient signal)

This is the first predictive theory for view selection in neural networks, filling a gap explicitly acknowledged in prior work. Our analysis suggests that the “dark knowledge” in soft labels is precisely the information about which view is present—information that hard labels cannot convey.

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A Proof of Theorem 1

[\[PLACEHOLDER: Full proof to be added when finalized\]](#)

B Proof of Theorem 2

[\[PLACEHOLDER: Full proof to be added when finalized\]](#)

C Proof of Theorem 3

[\[PLACEHOLDER: Full proof to be added when finalized\]](#)

D Additional Experimental Details

[\[PLACEHOLDER: Hyperparameters, training details, additional results\]](#)